
CE 556P: Structural Engineering Laboratory

Course Code	CE 556P
Course Name	Structural Engineering Laboratory
L-T-P-C	0-0-4-2
Prerequisites	NIL
Intended for	PG (MS, MTech, PhD)
Distribution	Specialization Core (MTech in Structural Engineering), Elective (MS, PhD)
Approval	23rd BoA (Apr 2018); 16th Senate (Apr 2018)

Course Contents

- **Self-Compacting and High-Strength Concrete:** [12 Hours]
 - Mix-design for self-compacting and high-strength concrete
 -) Stress-strain characterization of self-compacting and high-strength concrete
- **Behavior of Structural Elements:** [16 Hours]
 - Behavior of Reinforced Concrete (RC) beam under flexure and shear
 - Un-symmetrical bending of steel beam
 - Behavior of slab
- **Non-Destructive Testing and Damage Detection** [8 Hours]
- **Model Testing for Dynamic Characterization:** [20 Hours]
 - Free and forced vibrations of structure and evaluation of dynamic characteristics
 - Dynamic Young's modulus, shear modulus, and Poisson's ratio of materials
 - Behavior of frame and shear wall building models under horizontal excitation
 - Time and frequency-domain study for dynamic response analysis

Text Books

1. H.G. Harris, G. Sabnis, **Structural Modeling and Experimental Techniques**, 2nd Edition, CRC Press, 1999.

References

1. W.F. Sharpe, **Springer Handbook of Experimental Solid Mechanics**, Springer, 2008.
2. V.M. Malhotra, N.J. Carino, **Handbook of Nondestructive Testing of Concrete**, 2nd Edition, CRC Press, 2003.
3. R.E. Coleman, **Experimental Structural Dynamics: An Introduction to Experimental Methods of Characterizing Vibrating Structures**, AuthorHouse, 2004.